1.1

$N(14) = 1$ $\quad N(22) = 2$

$N(15) = 1$ $\quad N(24) = 2$

$N(16) = 3$ $\quad N(25) = 5$

(a) $\langle j \rangle = \dfrac{14\cdot 1 + 15\cdot 1 + 16\cdot 3 + 22\cdot 2 + 24\cdot 2 + 25\cdot 5}{14}$

$= \dfrac{1}{14}\left[14 + 15 + 48 + 44 + 48 + 125\right]$

$= \dfrac{294}{14} = 21$

$\Rightarrow \langle j \rangle^2 = 441$

$\langle j^2 \rangle = \dfrac{196\cdot 1 + 225\cdot 1 + 256\cdot 3 + 484\cdot 2 + 576\cdot 2 + 625\cdot 5}{14}$

$= \dfrac{6434}{14} = \dfrac{3217}{7}$

(e) $\Delta j = j - \langle j \rangle$

j	Δj
14	-7
15	-6
16	-5
22	1
24	3
25	4

$\Rightarrow \sigma^2 = \dfrac{(-7)^2 + (-6)^2 + (-5)^2\cdot 3 + 1\cdot 2 + 3\cdot 2 + 16\cdot 5}{14}$

$= \dfrac{260}{14} = \dfrac{130}{7}$

$\Rightarrow \sigma = \sqrt{\dfrac{130}{7}}$

(c) $\sigma = \sqrt{\dfrac{3217}{7} - 441} = \sqrt{\dfrac{130}{7}}$

1.2

(a) $f(x) = \frac{1}{2\sqrt{hx}} \quad (0 \le x \le h)$

$$\langle x \rangle^2 = \left(\frac{h}{3}\right)^2 = \frac{h^2}{9}$$

$$\langle x^2 \rangle = \int_0^h x^2 \frac{1}{2\sqrt{hx}}\,dx = \frac{1}{2\sqrt{h}} \frac{2}{5} h^{5/2} = \frac{h^2}{5}$$

$$\Rightarrow \sigma = \sqrt{\frac{h^2}{5} - \frac{h^2}{9}} = \sqrt{\frac{4h^2}{45}} = \frac{2}{3}\frac{h}{\sqrt{5}}$$

(b) $\text{Prob}(\text{more than } 1\sigma \text{ away}) = 1 - \int_{\langle x\rangle - \sigma}^{\langle x\rangle + \sigma} f(x)\,dx$

$$= 1 - \int_{\langle x\rangle - \sigma}^{\langle x\rangle + \sigma} \frac{1}{2\sqrt{hx}}\,dx = 1 - \frac{2}{2\sqrt{h}}\sqrt{x}\Big|_{\langle x\rangle - \sigma}^{\langle x\rangle + \sigma}$$

$$= 1 - \frac{1}{\sqrt{h}}\left[\left(\frac{h}{3} + \frac{h}{3}\frac{2}{\sqrt{5}}\right)^{1/2} - \left(\frac{h}{3} - \frac{h}{3}\frac{2}{\sqrt{5}}\right)^{1/2}\right]$$

$$= 1 - \frac{1}{\sqrt{3}}\left[\left(1 + \frac{2}{\sqrt{5}}\right)^{1/2} - \left(1 - \frac{2}{\sqrt{5}}\right)^{1/2}\right]$$

$$\approx 0.393$$

1.3 $\rho(x) = A e^{-\lambda(x-a)^2}$

(a) $\int_{-\infty}^{+\infty} A e^{-\lambda(x-a)^2} dx$ $\qquad \sqrt{\lambda}(x-a) = z \qquad \Rightarrow dx = \frac{dz}{\sqrt{\lambda}}$

$= \frac{1}{\sqrt{\lambda}} \int_{-\infty}^{+\infty} A e^{-z^2} dz = \frac{A\sqrt{\pi}}{\sqrt{\lambda}} \Rightarrow \boxed{A = \sqrt{\frac{\lambda}{\pi}}}$

(e) $\langle x \rangle = \int_{-\infty}^{+\infty} A x e^{-\lambda(x-a)^2}$ $\qquad x-a = y$

$= \int_{-\infty}^{+\infty} A(y+a) e^{-\lambda y^2} dy$

$= a A \sqrt{\frac{\pi}{\lambda}} = \boxed{a = \langle x \rangle}$

$\langle x^2 \rangle = \int_{-\infty}^{+\infty} A x^2 e^{-\lambda(x-a)^2} dx$ $\qquad \sqrt{\lambda}(x-a) = z$

$= A \frac{1}{\sqrt{\lambda}} \int_{-\infty}^{+\infty} \left(\frac{z}{\sqrt{\lambda}} + a\right)^2 e^{-z^2} dz$

$= A \frac{1}{\sqrt{\lambda}} \left[\int_{-\infty}^{+\infty} \frac{z^2}{\lambda} e^{-z^2} dz + a^2 \sqrt{\pi} \right]$

$= A \frac{1}{\sqrt{\lambda}} \left[a^2 \sqrt{\pi} + \frac{1}{\lambda} \underbrace{\int_{-\infty}^{+\infty} z^2 e^{-z^2} dz}_{I_2} \right]$

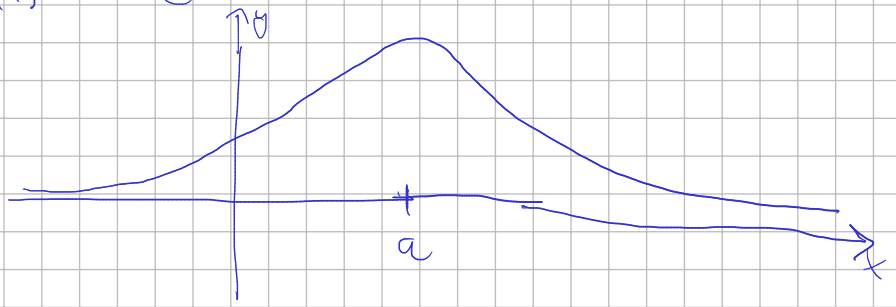
$I(\alpha) = \int_{-\infty}^{+\infty} e^{-\alpha z^2} dz \qquad \Rightarrow I_2 = -\frac{dI}{d\alpha}\Big|_{\alpha=1}$

$I(\alpha) = \sqrt{\frac{\pi}{\alpha}} \Rightarrow I_2 = -\frac{d}{d\alpha}\left[\sqrt{\frac{\pi}{\alpha}}\right]\Big|_{\alpha=1} = \frac{\sqrt{\pi}}{2}$

$\Rightarrow \langle x^2 \rangle = \frac{1}{\sqrt{\pi}}\left[a^2\sqrt{\pi} + \frac{1}{\lambda}\frac{\sqrt{\pi}}{2}\right] = \boxed{a^2 + \frac{1}{2\lambda} = \langle x^2 \rangle}$

$\Rightarrow \sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = a^2 + \frac{1}{2\lambda} - a^2 = \boxed{\frac{1}{2\lambda} = \sigma^2}$

(c) $\rho(x) = A e^{-\lambda|x-a|^2}$



1.4

$$\psi(x,0) = \begin{cases} A\frac{x}{a} & 0 \le x \le a \\ A\frac{b-x}{b-a} & a \le x \le b \\ 0 & \text{else} \end{cases} \qquad (b > a > 0)$$

(a)

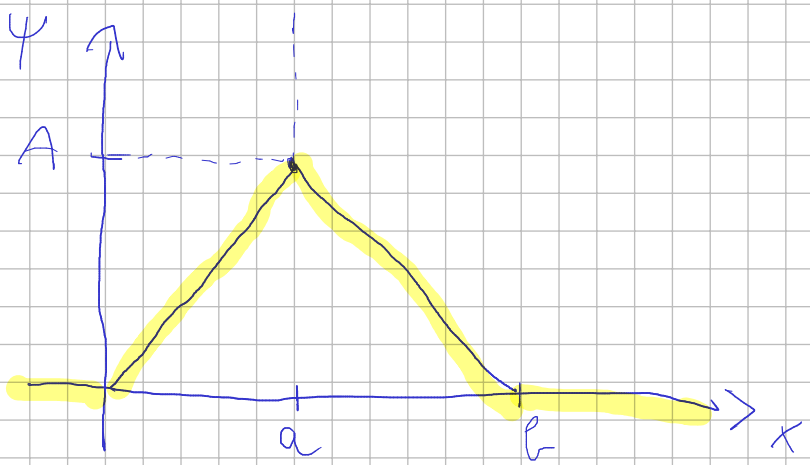
$$1 = \int_{-\infty}^{+\infty} |\psi|^2 dx = \frac{A^2}{a^2}\int_0^a x^2 dx + \frac{A^2}{(b-a)^2}\int_a^b (b-x)^2 dx$$

$b - x = y$

$$= \frac{A^2 a}{3} - \frac{A^2}{(b-a)^2}\int_{b-a}^0 y^2 dy = \frac{A^2 a}{3} + \frac{A^2(b-a)}{3} = \frac{A^2 b}{3}$$

$$\Rightarrow A = \sqrt{\frac{3}{b}}$$

(b)



(c) At $t=0$, most likely at $x=a$

(d) Left of a $\Rightarrow$ $\int_0^a |\psi|^2 dx = \frac{A^2}{a^2}\int_0^a x^2 dx = \frac{A^2 a}{3} = \boxed{\frac{a}{b}}$

If $b = a \Rightarrow P(\text{left of } a) = \frac{a}{a} = 1$ OK

If $b = 2a \Rightarrow P(\text{left of } a) = \frac{a}{2a} = \frac{1}{2}$ OK $\quad y = b - x$

(e) $$\langle x \rangle = \int_{-\infty}^{+\infty} x|\psi|^2 dx = \frac{3}{a^2 b}\int_0^a x^3 dx + \frac{3}{b(b-a)^2}\int_a^b x(b-x)^2 dx$$

$$= \frac{3a^2}{4b} - \frac{3}{b(b-a)^2}\int_{b-a}^0 (b-y)y^2 dy = \frac{3a^2}{4b} - \frac{3}{b(b-a)^2}\left[b\frac{y^3}{3} - \frac{y^4}{4}\right]_{b-a}^0$$

$$= \frac{3a^2}{4b} - \frac{3}{b(b-a)^2}\left[-\frac{b}{3}(b-a)^3 + \frac{1}{4}(b-a)^4\right] = \frac{3a^2}{4b} + b - a - \frac{3}{4b}(b^2 - 2ab + a^2)$$

$$= \frac{1}{4b}\left[3a^2 + 4b^2 - 4ab - 3b^2 + 6ab - 3a^2\right]$$

$$= \frac{1}{4b}\left[b^2 + 2ab\right] = \boxed{\frac{2a+b}{4}}$$

1.5

$$\psi(x,t) = A e^{-\lambda|x|} e^{-i\omega t}$$

(a)

$$1 = \int_{-\infty}^{+\infty} A^2 e^{-2\lambda|x|}\,dx = 2A^2 \int_0^{+\infty} e^{-2\lambda x}\,dx$$

$$= 2A^2 \left.\frac{e^{-2\lambda x}}{-2\lambda}\right|_0^{+\infty} = \frac{A^2}{\lambda} \Rightarrow \boxed{A = \sqrt{\lambda}}$$

(b)

$$\langle x \rangle = \int_{-\infty}^{+\infty} \lambda e^{-2\lambda|x|}\, x\,dx = \boxed{0 = \langle x \rangle}$$

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} \lambda e^{-2\lambda|x|}\, x^2\,dx = 2\lambda \int_0^{+\infty} e^{-2\lambda x}\, x^2\,dx$$

$$= 2\lambda \left[\left.\frac{e^{-2\lambda x}}{-2\lambda} x^2\right|_0^{+\infty} + \frac{1}{\lambda}\int_0^{+\infty} e^{-2\lambda x}\, x\,dx \right]$$

$$= 2\frac{1}{2\lambda}\int_0^{+\infty} e^{-2\lambda x}\,dx = \frac{1}{\lambda}\left[\frac{e^{-2\lambda x}}{-2\lambda}\right]_0^{+\infty}$$

$$\Rightarrow \boxed{\langle x^2 \rangle = \frac{1}{2\lambda^2}}$$

(c)

$$\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{1}{2\lambda^2}$$

$$\Rightarrow \boxed{\sigma = \frac{1}{\sqrt{2}\,\lambda}}$$

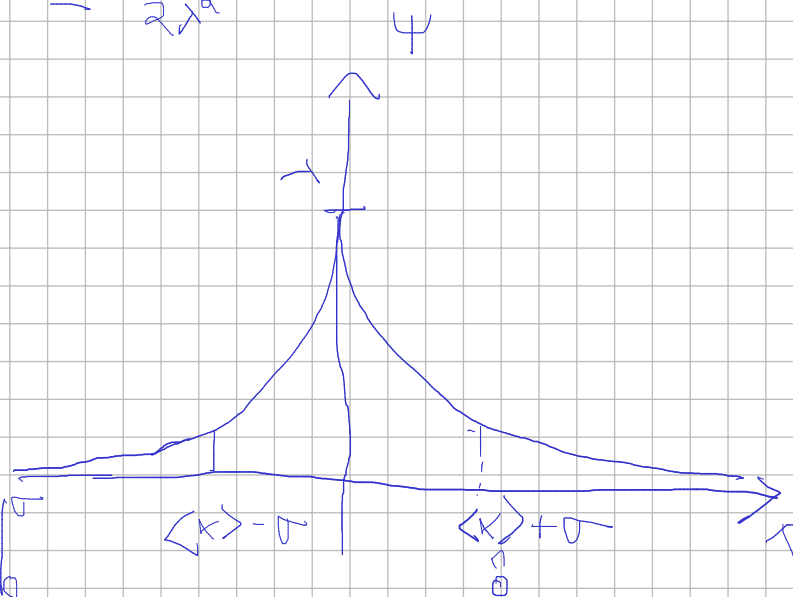
$$|\psi|^2 = \lambda e^{-2\lambda|x|}$$

$$P(-\sigma < x < \sigma) = \int_{-\sigma}^{+\sigma} \lambda e^{-2\lambda|x|}\,dx$$

$$= 2\lambda \int_0^{\sigma} e^{-2\lambda x}\,dx = 2\lambda \left.\frac{e^{-2\lambda x}}{-2\lambda}\right|_0^{\sigma}$$

$$= 1 - e^{-2\lambda\sigma}$$

$$\Rightarrow P(\text{outside } [-\sigma,\sigma]) = 1 - \left(1 - e^{-2\lambda\sigma}\right) = e^{-2\lambda\sigma} = e^{-\frac{2\lambda}{\sqrt{2}\lambda}} = e^{-\sqrt{2}}$$



1.6

The integral is over x, not over t.

1.7 $\langle p \rangle = \int \psi^* \left[-i\hbar \frac{\partial}{\partial x} \right] \psi \, dx$

$\frac{\partial \psi}{\partial t} = \frac{i\hbar}{2m} \frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V\psi \quad \| \quad \frac{\partial \psi^*}{\partial t} = -\frac{i\hbar}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar} V\psi^*$

$\Rightarrow \frac{d\langle p \rangle}{dt} = \int \frac{\partial \psi^*}{\partial t} \left[-i\hbar \frac{\partial}{\partial x} \right] \psi \, dx + \int \psi^* \left[-i\hbar \frac{\partial}{\partial x} \right] \frac{\partial \psi}{\partial t} dx$

$= \int \left(-\frac{i\hbar}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar} V\psi^* \right) \left[-i\hbar \frac{\partial}{\partial x} \right] \psi \, dx$

$+ \int \psi^* \left[-i\hbar \frac{\partial}{\partial x} \right] \left(\frac{i\hbar}{2m} \frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V\psi \right) dx$

$= -\frac{\hbar^2}{2m} \int \frac{\partial^2 \psi^*}{\partial x^2} \frac{\partial \psi}{\partial x} dx + \frac{\hbar^2}{2m} \int \psi^* \frac{\partial^3 \psi}{\partial x^3} dx + \int \psi^* V \frac{\partial \psi}{\partial x} dx - \int \psi^* \frac{\partial}{\partial x}(V\psi) \, dx$

$\circ$ by integration by parts

$= -\int \psi^* \frac{\partial V}{\partial x} \psi \, dx = \left\langle -\frac{dV}{dx} \right\rangle$

1.8 $i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi + V_0 \psi$ with $V_0 = \text{CONSTANT}$

Let ψ_0 be the solution of the equation with $V_1 = 0$

Let $\psi_1 = \psi_0 \, e^{-\frac{i}{\hbar} V_0 t}$

$\Rightarrow i\hbar \frac{\partial \psi_1}{\partial t} = i\hbar \frac{\partial \psi_0}{\partial t} e^{-\frac{i}{\hbar} V_0 t} + i\hbar \psi_0 \left(-\frac{i V_0}{\hbar} \right) e^{-\frac{i}{\hbar} V_0 t}$

$= \left[-\frac{\hbar^2}{2m} \frac{\partial^2 \psi_0}{\partial x^2} + V\psi_0 \right] e^{-\frac{i}{\hbar} V_0 t} + V_0 e^{-\frac{i}{\hbar} V_0 t}$ QED

No effect on exp. value (as long as operator does not include time derivative), as $|\psi|^2$ is unaffected by phase.

7.9 $\psi(x,t) = A e^{-a\left[\frac{mx^2}{\hbar} + it\right]}$

(a) $1 = \int A^2 e^{-a\left[\frac{mx^2}{\hbar} + it\right]} e^{-a\left[\frac{mx^2}{\hbar} - it\right]} dx$

$= A^2 \int e^{-\frac{2am}{\hbar}x^2} dx = A^2 \sqrt{\frac{\pi\hbar}{2am}}$

$\Rightarrow \boxed{A = \left[\frac{2am}{\pi\hbar}\right]^{1/4}}$

(b) $i\hbar \frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi$ (SE)

$\frac{\partial\psi}{\partial t} = -ai\psi$ $\qquad \frac{\partial\psi}{\partial x} = -\frac{2amx}{\hbar}\psi$

$\frac{\partial^2\psi}{\partial x^2} = -\frac{2am}{\hbar}\psi + \frac{4a^2m^2x^2}{\hbar^2}\psi$

(SE) $\Rightarrow a\hbar = -\frac{\hbar^2}{2m}\left[-\frac{2am}{\hbar} + \frac{4a^2m^2x^2}{\hbar^2}\right] + V$

$\Rightarrow \boxed{V = 2a^2mx^2}$

(c) $\langle x\rangle = \int \psi^* x \psi\, dx = 0$

$\langle x^2\rangle = \int A^2 e^{-\frac{2amx^2}{\hbar}} x^2 dx$

$\sqrt{\frac{2am}{\hbar}}x = y$

$= A^2\left(\frac{\hbar}{2am}\right)^{3/2}\int e^{-y^2} y^2 dy$

$u = y$

$dv = ye^{-y^2}dy \Rightarrow v = -\frac{e^{-y^2}}{2}$

$= A^2\left(\frac{\hbar}{2am}\right)^{3/2}\cdot\frac{1}{2}\int e^{-y^2}dy = \left[\frac{2am}{\pi\hbar}\right]^{1/2}\left(\frac{\hbar}{2am}\right)^{3/2}\cdot\frac{\sqrt{\pi}}{2}$

$\boxed{\langle x^2\rangle = \frac{\hbar}{4ma}}$

$\psi = Ae^{-a\left[\frac{mx^2}{\hbar}\right]}$

$\langle p\rangle = \int \psi^*\left(-i\hbar\frac{\partial}{\partial x}\right)\psi = -i\hbar A^2\int e^{-\frac{2amx^2}{\hbar}}\left(-\frac{2amx}{\hbar}\right)dx = 0$

$\langle p^2\rangle = \int \psi^*\left(-i\hbar\frac{\partial}{\partial x}\right)^2\psi = -\hbar^2A^2\int e^{-\frac{amx^2}{\hbar}}\frac{\partial}{\partial x}\left(e^{-a\frac{mx^2}{\hbar}}\cdot\left(-\frac{2amx}{\hbar}\right)\right)dx$

$= \hbar A^2 2am\left[\int e^{-\frac{2amx^2}{\hbar}}\left(-\frac{2amx^2}{\hbar}\right) + \sqrt{\frac{\pi\hbar}{2am}}\right]$

$$= \hbar A^2 2am \sqrt{\frac{\pi\hbar}{2am}} - 4A^2a^2m^2 \int e^{-\frac{2amx^2}{\hbar}} x^2\, dx$$

$$= \hbar A^2 2am \sqrt{\frac{\pi\hbar}{2am}} - 4A^2a^2m^2 \left(\frac{\hbar}{2am}\right)^{3/2} \frac{\sqrt{\pi}}{2}$$

$$= \frac{1}{2}A^2 \sqrt{2\pi am}\, \hbar^{3/2} = \frac{1}{2}\left[\frac{2am}{\pi\hbar}\right]^{1/2} \sqrt{2\pi am}\, \hbar^{3/2}$$

$$\Rightarrow \boxed{\langle p^2 \rangle = a\hbar m}$$

(d) $\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{\hbar}{4ma}$

$\sigma_p = \langle p^2 \rangle - \langle p \rangle^2 = a\hbar m$

$$\Rightarrow \sigma_x \sigma_p = \sqrt{\frac{\hbar^2}{4}} = \frac{\hbar}{2} \geq \frac{\hbar}{2} \Rightarrow$$ Uncertainty OK!

1.10

3.14159265358979323846264338327950288419716939937510

(a) Prob column →

(b) most probable = 3

$P(x<4) = \frac{10}{25}$

$P(x<5) = \frac{13}{25}$ ⇒ median = 4

	#	Prob	$(j - \langle j \rangle) \cdot 25$
0	0	0	-118
1	2	2/25	-93
2	3	3/25	-68
3	5	5/25	-43
4	3	3/25	-18
5	3	3/25	7
6	3	3/25	32
7	1	1/25	57
8	2	2/25	82
9	3	3/25	107

Average $= \frac{1}{25}\left[1\cdot 2 + 2\cdot 3 + 3\cdot 5 + 4\cdot 3 + 5\cdot 3 + 6\cdot 3 + 7\cdot 1 + 8\cdot 2 + 9\cdot 3\right] = \frac{118}{25}$

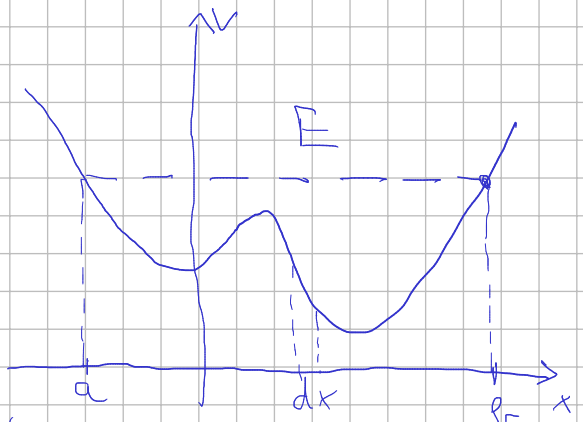
(d) $\sigma^2 = \frac{1}{(25)^3}\left[2\cdot(93)^2 + 3\cdot(68)^2 + 5\cdot(43)^2 + 3\cdot(18)^2 + 3\cdot(7)^2 + 3\cdot(32)^2 + 1\cdot(57)^2 + 2\cdot(82)^2 + 3\cdot(107)^2\right]$

$\Rightarrow \sigma \approx 2.474$

1.11

$$v = \frac{dx}{dt}$$

$$T = \int_a^b \frac{dx}{v(x)} \qquad \rho(x) = \frac{1}{v(x)\,T}$$



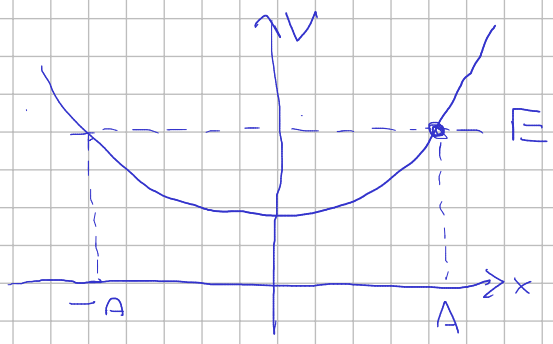
(a) $E = \frac{1}{2} m v^2(x) + V(x)$

$$\Rightarrow v(x) = \sqrt{\frac{2\left[E - V(x)\right]}{m}}$$

(v is taken to be the absolute value of the velocity, so only positive root)

(b) $V(x) = \frac{1}{2} k x^2$

$$\Rightarrow v(x) = \left[\frac{2}{m}\left(E - \frac{1}{2}kx^2\right)\right]^{1/2}$$



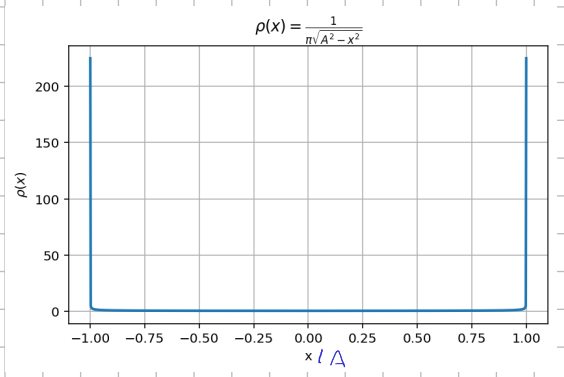
$$T = \int_{-A}^{A} \frac{dx}{\left[\frac{2}{m}\left(\frac{1}{2}kA^2 - \frac{1}{2}kx^2\right)\right]^{1/2}}$$

$$= \sqrt{\frac{m}{k}} \frac{1}{A} \int_{-A}^{A} \frac{dx}{\sqrt{1 - \left(\frac{x}{A}\right)^2}} \qquad \frac{x}{A} = y$$

$$= \sqrt{\frac{m}{k}} \int_{-1}^{1} \frac{dy}{\sqrt{1-y^2}} = \sqrt{\frac{m}{k}} \left[A\sin(1) - A\sin(-1)\right] = \pi\sqrt{\frac{m}{k}}$$

$$T\,v(x) = \pi\sqrt{\frac{m}{k}}\left[\frac{2}{m}\left(\frac{1}{2}kA^2 - \frac{1}{2}kx^2\right)\right]^{1/2} = \pi\sqrt{A^2 - x^2} = \pi\sqrt{\frac{2E}{k} - x^2}$$

$$\Rightarrow \rho(x) = \frac{1}{\pi\sqrt{A^2 - x^2}}$$



$$\int_{-A}^{A} \frac{dx}{\pi\sqrt{A^2 - x^2}} \qquad \frac{x}{A} = y$$

$$= \frac{1}{\pi} \int_{-1}^{1} \frac{dy}{\sqrt{1-y^2}} = 1$$

(c) $$\langle x \rangle = \int_{-A}^{A} x\,\rho(x)\,dx = \int_{-A}^{A} \frac{x\,dx}{\pi\sqrt{A^2 - x^2}} = 0$$

$$\langle x^2 \rangle = \int_{-A}^{A} \frac{x^2 \, dx}{\pi \sqrt{A^2 - x^2}}$$

$$\frac{x}{A} = \cos\theta \qquad \left| \cos(2\alpha) = 2\cos^2\alpha - 1 \right.$$

$$\Rightarrow \cos^2\alpha = \frac{1 + \cos(2\alpha)}{2}$$

$$= -\frac{A^2}{\pi} \int_{\pi}^{0} \frac{\cos^2\theta \, \sin\theta \, d\theta}{\sin\theta} = -\frac{A^2}{\pi} \int_{\pi}^{0} \frac{1 + \cos(2\theta)}{2} \, d\theta$$

$$= -\frac{A^2}{2\pi} \left[-\pi + \frac{\sin 2\theta}{2} \Big|_{-\pi}^{0} \right] = \frac{A^2}{2}$$

$$\Rightarrow \sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{A^2}{2}$$

$$\Rightarrow \sigma_x = \frac{A}{\sqrt{2}} = \sqrt{\frac{E}{k}}$$

1.12

$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$

At $x_{max} = A$, we have $v = 0$

At $x = 0$, we have $E = \frac{1}{2}mv_{max}^2$

$\Rightarrow v \in \left[-\sqrt{\frac{2E}{m}}, \sqrt{\frac{2E}{m}}\right] \Rightarrow p = mv \in \left[-\sqrt{2mE}, \sqrt{2mE}\right]$

(a) $E = \frac{p^2}{2m} + \frac{1}{2}kx^2 \Rightarrow x(p) = \left[\frac{2mE - p^2}{mk}\right]^{1/2}$

$\rho(x)\,dx = \rho(x(p))\left|\frac{dx}{dp}\right| dp$

$\Rightarrow \varphi(p) = \rho(x(p))\left|\frac{dx}{dp}\right| = \frac{1}{\pi\sqrt{\frac{2E}{k} - x^2(p)}} \cdot \frac{1}{2}\left[\frac{2mE - p^2}{mk}\right]^{-\frac{1}{2}} \cdot \frac{2p}{mk}$

$= \frac{p}{\pi\sqrt{mk}}\left[\frac{2mE}{mk} - \frac{2mE - p^2}{mk}\right]^{-1/2} \left[2mE - p^2\right]^{-\frac{1}{2}}$

$= \frac{p}{\pi} \cdot p^{-1}\left[2mE - p^2\right]^{-\frac{1}{2}}$

$$\boxed{\varphi(p) = \frac{1}{\pi\sqrt{2mE - p^2}}}$$

(b) $\langle p \rangle = \int_{-\sqrt{2mE}}^{\sqrt{2mE}} p\,\varphi(p)\,dp = 0$ (because φ is even in p)

$\langle p^2 \rangle = \int_{-\sqrt{2mE}}^{\sqrt{2mE}} \frac{p^2}{\pi\sqrt{2mE - p^2}}\,dp = \frac{1}{\pi\sqrt{2mE}} \int_{-\sqrt{2mE}}^{\sqrt{2mE}} \frac{p^2}{\sqrt{1 - \left(\frac{p}{\sqrt{2mE}}\right)^2}}\,dp$

$= \frac{2mE \cdot \sqrt{2mE}}{\pi\sqrt{2mE}} \int_{-1}^{1} \frac{z^2}{\sqrt{1 - z^2}}\,dz$ $\qquad \frac{p}{\sqrt{2mE}} = z$

$z = \cos\theta$

$= -\frac{2mE}{\pi} \int_{-\pi}^{0} \frac{\cos^2\theta \sin\theta}{-\sin\theta}\,d\theta = \frac{2mE}{\pi} \int_{-\pi}^{0} \frac{1 + \cos 2\theta}{2}\,d\theta$

$= \frac{2mE}{\pi}\left[\frac{\pi}{2} + 0\right]$

$= mE$

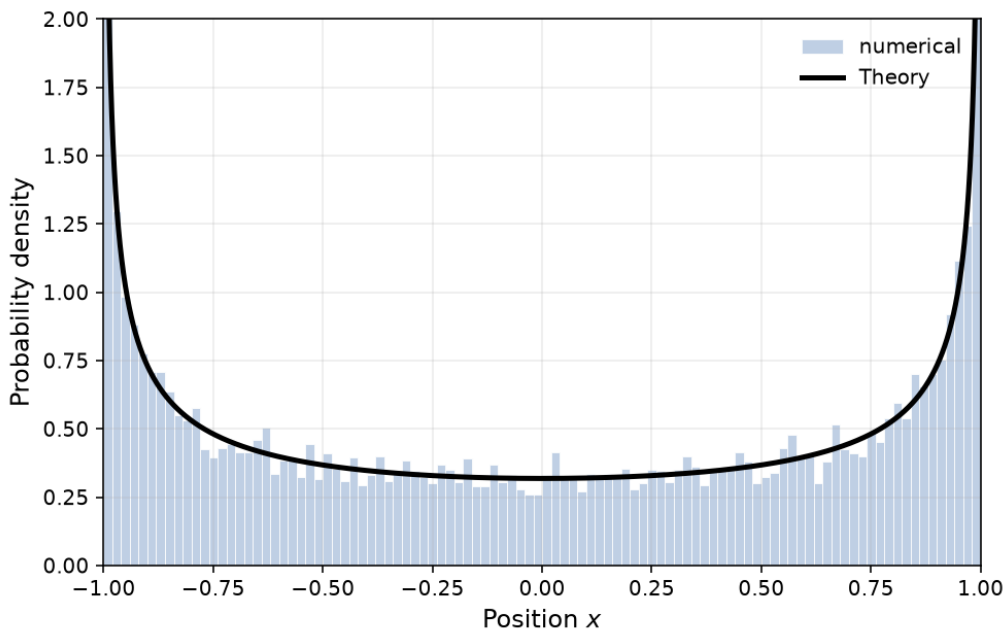
$\Rightarrow \sigma_p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{mE}$

(c) $\sigma_x \sigma_p = \sqrt{\frac{E}{k}} \cdot \sqrt{mE} = \sqrt{\frac{m}{k}} E$

If $E \geq \frac{\hbar\omega}{2} = \frac{\hbar}{2}\sqrt{\frac{k}{m}}$

$\Rightarrow \sigma_x \sigma_p \geq \frac{\hbar}{2}$

1.13



Code on github

1.14

$$\frac{dP_{a,b}}{dt} = \frac{d}{dt}\int_a^b |\psi(x,t)|^2 dx = \int_a^b \left[\frac{\partial \psi^*}{\partial t}\cdot\psi + \psi^*\frac{\partial \psi}{\partial t}\right]dx$$

$$= \int_a^b \left[\psi\left(-\frac{i\hbar}{2m}\frac{\partial^2\psi^*}{\partial x^2} + \frac{i}{\hbar}V\psi^*\right) + \psi^*\left(\frac{i\hbar}{2m}\frac{\partial^2\psi}{\partial x^2} - \frac{i}{\hbar}V\psi\right)\right]dx$$

$$= -\frac{i\hbar}{2m}\left[\psi\frac{\partial\psi^*}{\partial x}\Big|_a^b - \int_a^b \frac{\partial\psi}{\partial x}\frac{\partial\psi^*}{\partial x}dx\right]$$

$$+ \frac{i\hbar}{2m}\left[\psi^*\frac{\partial\psi}{\partial x}\Big|_a^b - \int_a^b \frac{\partial\psi}{\partial x}\frac{\partial\psi^*}{\partial x}dx\right]$$

$$= \frac{i\hbar}{2m}\left[\psi(a,t)\frac{\partial\psi^*(a,t)}{\partial x} - \psi^*(a,t)\frac{\partial\psi(a,t)}{\partial x}\right]$$

$$- \frac{i\hbar}{2m}\left[\psi(b,t)\frac{\partial\psi^*(b,t)}{\partial x} - \psi^*(b,t)\frac{\partial\psi(b,t)}{\partial x}\right]$$

$$= J(a,t) - J(b,t)$$

1.15

$$\frac{d}{dt}\int_{-\infty}^{+\infty}\psi_1^*\psi_2\,dx = \int_{-\infty}^{+\infty}\left[\frac{\partial\psi_1^*}{\partial t}\psi_2 + \psi_1^*\frac{\partial\psi_2}{\partial t}\right]dx$$

$$= \int_{-\infty}^{+\infty}\left[\psi_2\left(\frac{-i\hbar}{2m}\frac{\partial^2\psi_1^*}{\partial x^2} + \frac{i}{\hbar}V\psi_1^*\right) + \psi_1^*\left(\frac{i\hbar}{2m}\frac{\partial^2\psi_2}{\partial x^2} - \frac{i}{\hbar}V\psi_2\right)\right]dx$$

$$= \frac{i\hbar}{2m}\left[\psi_1^*\frac{\partial\psi_2}{\partial x}\Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty}\frac{\partial\psi_1^*}{\partial x}\frac{\partial\psi_2}{\partial x}dx - \psi_2\frac{\partial\psi_1^*}{\partial x}\Big|_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty}\frac{\partial\psi_1^*}{\partial x}\frac{\partial\psi_2}{\partial x}dx\right]$$

$$= 0$$

1. 16

(a) $1 = \int_{-a}^{a} A^2 (a^2 - x^2)^2 \, dx = A^2 \left[a^4 x - \frac{2}{3} a^2 x^3 + \frac{1}{5} x^5 \right]_{-a}^{a}$

$= 2A^2 \left[a^5 - \frac{2}{3} a^5 + \frac{1}{5} a^5 \right] = \frac{2A^2}{15} \left[15 - 10 + 3 \right] a^5$

$= \frac{16}{15} a^5 A^2$

$\Rightarrow \boxed{A = \sqrt{\frac{15}{16 a^5}}}$

(b) $\langle x \rangle = \int_{-a}^{a} A^2 (x^2 - a^2)^2 \, x \, dx = \boxed{0}$

(c) For $\langle p \rangle = m \frac{d\langle x \rangle}{dt}$ we need $\psi(x,t)$. But only $\psi(x,0)$ is given. So we cannot take a time derivative.

$\langle p \rangle = \int_{-a}^{a} \psi^* \left[-i\hbar \frac{\partial}{\partial x} \right] \psi \, dx$

$= -i\hbar A^2 \int_{-a}^{a} (a^2 - x^2)(-2x) \, dx = \boxed{0}$

(d) $\langle x^2 \rangle = \int_{-a}^{a} A^2 (a^2 - x^2)^2 x^2 \, dx = \frac{15}{16 a^5} \left[\frac{a^4}{3} x^3 - \frac{2a^2}{5} x^5 + \frac{x^7}{7} \right]_{-a}^{a}$

$= \frac{15}{8} a^2 \frac{35 - 42 + 15}{105} = \boxed{\frac{a^2}{7}}$

(e) $\langle p^2 \rangle = \int_{-a}^{a} \psi^* \left(-\hbar^2 \frac{\partial^2}{\partial x^2} \right) \psi \, dx$

$= -\hbar^2 A^2 \int_{-a}^{a} (a^2 - x^2) \cdot (-2) \, dx = 2\hbar^2 \frac{15}{16 a^5} \cdot \left[a^2 x - \frac{x^3}{3} \right]_{-a}^{a}$

$= \frac{15}{8} \frac{\hbar^2}{a^5} \frac{4}{3} a^3 = \boxed{\frac{5}{2} \frac{\hbar^2}{a^2}}$

(f) $\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{a^2}{7} - 0$

$\Rightarrow \boxed{\sigma_x = \frac{a}{\sqrt{7}}}$

(g) $\sigma_p^2 = \langle p^2 \rangle - \langle p \rangle^2 = \frac{5}{2} \frac{\hbar^2}{a^2} - 0$

$\Rightarrow \boxed{\sigma_p = \sqrt{\frac{5}{2}} \frac{\hbar}{a}}$

(h) $\sigma_x \sigma_p = \sqrt{\frac{5}{14}}\, \hbar = \sqrt{\frac{10}{7}} \frac{\hbar}{2} > \frac{\hbar}{2} \Rightarrow$ OK

1.17 $V = V_0 - i\Gamma$

(a) $\frac{\partial \psi}{\partial t} = \frac{i\hbar}{2m}\frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V \psi \Rightarrow \frac{\partial \psi^*}{\partial t} = -\frac{i\hbar}{2m}\frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar} V^* \psi^*$

$$\frac{\partial P}{\partial t} = \int_{-\infty}^{+\infty} \left[\frac{\partial \psi}{\partial t}\cdot \psi^* + \psi \frac{\partial \psi^*}{\partial t}\right] dx$$

$$= \int_{-\infty}^{+\infty} \left[\psi^* \left(\frac{i\hbar}{2m}\frac{\partial^2 \psi}{\partial x^2} - \frac{i}{\hbar} V_0 \psi - \frac{\Gamma}{\hbar}\psi\right) + \psi\left(-\frac{i\hbar}{2m}\frac{\partial^2 \psi^*}{\partial x^2} + \frac{i}{\hbar} V_0 \psi^* - \frac{\Gamma}{\hbar}\psi^*\right)\right] dx$$

$$= \boxed{-\frac{2\Gamma}{\hbar} P}$$

(b) $P = P_0 e^{-\frac{2\Gamma}{\hbar} t}$ with $P_0 = 1$

$\Rightarrow P = e^{-\frac{2\Gamma}{\hbar} t}$

$\lambda = \frac{\ln 2}{T_{1/2}}$

$\Rightarrow \boxed{T_{1/2} = \frac{\hbar \cdot \ln 2}{2\Gamma}}$

7.18

Let d = characteristic size of system.

If $\lambda << d$, then a (classical) point particle description works well. If $\lambda >> d$, then the wave-like properties of the object cannot be neglected.

$\Rightarrow$ Non-classical behaviour if $\lambda = \frac{h}{p} > d$

$$\frac{p^2}{2m} = \frac{3}{2} k_B T \iff p = \sqrt{3 m k_B T}$$

$$\frac{h}{p} = \frac{h}{\sqrt{3 m k_B T}} > d \iff T < \frac{h^2}{3 m k_B d^2}$$

(a) If $\lambda_{electron} >$ lattice spacing, then we can't consider the electron as a localised point particle.

$$\Rightarrow T < \frac{(2\pi \times 1.05 \cdot 10^{-34})^2}{3 \times 9.1 \cdot 10^{-31} \times 1.38 \cdot 10^{-23} \times (3 \cdot 10^{-10})^2}$$

$$= \frac{43.52 \cdot 10^{-68}}{339.066 \cdot 10^{-74}} = 0.13 \cdot 10^6 = 130\,000\,K$$

$\Rightarrow$ QM description required at room temperature.

For the nucleus, if $\lambda_{nucleus} > d$, then its wave-like nature is non-negligible.

$$T < \frac{h^2}{3 m k_B d^2} = \frac{43.52 \cdot 10^{-68}}{3 \times 4.66 \cdot 10^{-26} \times 1.38 \cdot 10^{-23} \times (3 \cdot 10^{-10})^2}$$

$= 2.5\ K \Rightarrow$ At room temperature, behaves like point particle.

(c) $PV = N k_B T$

$V \sim d^3$. For 1 particle : $P d^3 = k_B T$

$$\Rightarrow d = \left[\frac{k_B T}{P} \right]^{1/3}$$

$$T < \frac{\hbar^2}{3 m k_B d^2} = \frac{\hbar^2 P^{2/3}}{3 m k_B^{5/3} T^{2/3}}$$

$$\Rightarrow T < \frac{\hbar^{6/5} P^{2/5}}{(3m)^{3/5} k_B}$$

$m_{He} = 6.64 \cdot 10^{-27}$ kg

For atmospheric conditions : $T < \frac{(2\pi \times 1.05 \cdot 10^{-34})^{6/5} \cdot (10^5)^{2/5}}{(3 \times 6.64 \cdot 10^{-27})^{3/5} \times 1.38 \cdot 10^{-23}}$

$$\Rightarrow T < \frac{1.52 \times 10^{-38}}{5.24 \times 10^{-39}} = 2.9 \text{ K}$$

$\Rightarrow$ Classical at room temperature.

$m_H = 1.7 \cdot 10^{-27}$ kg

$$T < \frac{\hbar^2}{3 m k_B d^2} = \frac{(2\pi \times 1.05 \times 10^{-34})^2}{3 \times 1.7 \times 10^{-27} \times 1.38 \times 10^{-23} \times (10^{-2})^2}$$

$$= \frac{4.35 \times 10^{-67}}{7.04 \times 10^{-54}} = 6.2 \times 10^{-14} \text{ K}$$

$\Rightarrow$ Classical at 3 K